THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2241 Introduction to Mathematical Analysis
Test 2 Report

Points	0–11	12–15	16–20	21–25	26–40
No. of students	4	6	7	7	7

Question	1(a)	1(b)	
Mean	2.65	2.16	
Question	2		
Mean	3.35		
Question	3		
Mean	3.71		
Question	4		
Mean	0.68		
Question	5(a)	5(b)	5(c)
Mean	3.13	1.74	2.26

Mean: 19.68

Highest: 33

- (Question 1a) The performance in this question is good. However, there are some subtle mistakes found in some of the work. When we apply the comparison test, we need to compare $\left|\dfrac{\tan(\sin n)}{n^3}\right|$ with $\dfrac{1}{n^3}$. The absolute value cannot be removed since $\tan(\sin n)$ takes up both positive and negative values infinitely often. In some examples of application of the comparison test, the absolute value is excluded only because all the terms are nonnegative in those examples.

  Another mistake is that some students regard $\tan 1$ as $\tan 1^\circ$, hence using an incorrect bound for $\tan 1$. In fact, in all advanced mathematics, the input of all trigonometric functions are real numbers, so radian is used instead of degree. But this does not affect the proof too much.

- (Question 1b) There are various attempts to compare the given series with another one. Quite a lot of elementary mistakes are found when handling inequalities and the logarithm function, such as writing $\ln(ab) = (\ln a)(\ln b)$, $\ln(3n^n) = n\ln(3n)$, etc. Although these are some basic mathematical concepts which are not the main scope of this course, making such errors might probably lead to a serious impact and some wrong deduction.

It is not easy to judge whether the series converges or not at first glance, unless you are familiar with such examples. One suggestion is to approximate the terms by picking out those dominating terms. For this series, we may get something like

$$\frac{1}{\ln(2n^n+1)} \approx \frac{1}{\ln(2n^n)} = \frac{1}{\ln 2 + n\ln n} \approx \frac{1}{n\ln n}.$$

Since the series $\sum_{n=2}^{\infty} \frac{1}{n\ln n}$ diverges, the given series should also diverge. This also suggests the difficulty of this question because the series $\sum_{n=2}^{\infty} \frac{1}{n\ln n}$ diverges slowly. But this is a useful result that you should keep in mind. In fact, Q1(a) of revision class 2 is of similar style and involves the same divergent series.

- (Question 2) The performance in this question is poor. It is regarded as a standard example in this course about continuity. We have seen many examples of this kind, such as Examples 4.15, 4.16 (i.e. Q3 of Tutorial 8), Q7(d) of Tutorial 7 and Q2 of Tutorial 8. I actually expect you should have sufficient practices on this kind of question and find it routine.

  Many students know that we should equate x^2-1 and $-\frac{(x+1)(3x+1)}{x}$ to find the points of continuity. However, saying f is continuous at a if and only if $a^2-1 = -\frac{(a+1)(3a+1)}{a}$ is not acceptable. Indeed, neither the 'if part' nor the 'only if' part is considered as obvious. This requires a result like the following: $f(x) = \begin{cases} g(x) & \text{if } x \in \mathbb{Q}, \\ h(x) & \text{if } x \notin \mathbb{Q} \end{cases}$ is continuous at a if and only if $\lim_{x\to a} g(x) = \lim_{x\to a} h(x) = f(a)$. The result is correct, and can be proved in a similar way as in the solution. However, using this stronger result is considered as trivializing the question, and hence is not allowed.

  The above result may look obvious intuitively. It is similar to $f(x) = \begin{cases} g(x) & \text{if } x \geqslant 0, \\ h(x) & \text{if } x < 0 \end{cases}$, where checking the continuity at 0 is the same as showing $\lim_{x\to 0^+} g(x) = \lim_{x\to 0^-} h(x) = f(0)$. However, it is quite important for the sets $\mathbb{Q}$ and $\mathbb{R}-\mathbb{Q}$ to be dense in $\mathbb{R}$ in order that the result holds. The following is a counterexample.

  Let $f(x) = \begin{cases} x & \text{if } x \in \mathbb{Z}, \\ x^2 & \text{if } x \notin \mathbb{Z} \end{cases}$. If we simply solve the equation $x = x^2$, we only get $x = 0, 1$, which are continuous points. However, apart from these points, f is also continuous at all non-integer points (why?).

  Therefore, when we say that we choose a sequence of rational/irrational numbers converging to a, bear in mind that this can be done because of the denseness of $\mathbb{Q}$ and $\mathbb{R}-\mathbb{Q}$.

- (Question 3) There are several ways to solve this question.

  The first suggested solution is to use the definition of uniform continuity. Most students following this method knew the correct approach but got stuck in the part of obtaining an upper bound for $\left|\frac{2xy+y+x-4}{(x^2+2)(y^2+2)}\right|$. This is certainly the tricky part of this question, but once again knowing how to obtain bounds for expressions is important in analysis.

In fact, the steps involved here are similar to those questions about contractive sequences. There was a similar expression in Q2 of Assignment 3, which requires us to obtain an upper bound for $\frac{x+y}{(x^2+2)(y^2+2)}$. Although this is a question in Section 2, I thought you should remember that because you should have worked on the assignment questions!

There is no general way to obtain a bound for some expression. Calculus may allow us to obtain the best bound easily in some cases, but this is not needed because we just need an upper bound here, which needs not be tight. If the expression goes to 0 when x or y goes to infinity, then normally some weak bounds already suffice. Also, applying the triangle inequality as in the solution may also simplify the things because we only need to focus on each individual term.

The second solution is to apply the result given in Q7 of revision class 2, i.e. it suffices to prove that g is uniformly continuous on each of the intervals $(-\infty, -M]$, $[-M, M]$ and $[M, \infty)$, where M is a sufficiently large number. By using the uniform continuity theorem, we easily know that g is uniformly continuous on $[-M, M]$. For the other two intervals, since we have the extra conditions $|x| \geqslant M$, the bounds can be much easier to obtain. Indeed, since $g(x) \to 0$ when $x \to \infty$, one can choose a sufficiently large M so that $|g(x)| < \varepsilon$ for $|x| \geqslant M$.

A third solution is to apply the result of Q2 in the latest assignment (which was not given before the test). Indeed, if we can show that g' is bounded, then we can prove that g is Lipschitz continuous, and hence uniformly continuous.

- (Question 4) This is a difficult question. The condition looks complicated. In fact, it suffices to have a weaker condition that for any $x \in (a, b)$, there exists $y > x$ such that $y \in [a, b]$ and $f(y) = f(x)$. The same proof as in the solution still works.

  The key idea of the solution comes from the intuitive idea as follows. For a certain x_1, we can find some $x_2 > x_1$ such that $f(x_2) = f(x_1)$. Then we find some $x_3 > x_2$ such that $f(x_3) = f(x_2) = f(x_1)$. This process continues indefinitely unless we reach the endpoint b. However, the trouble is that the process needs not stop. There maybe some strange functions like $f(x) = \sin \frac{1}{x}$ (for $x \neq 0$) which takes up the same value for an infinite number of times when x approaches to 0 (but there is a trouble of continuity at the point $x = 0$ no matter how $f(0)$ is defined). We need to find a rigorous way to reject such cases. Considering the 'largest' point with image equal to $f(x_1)$ seems to be the correct idea, but this 'largest' point needs not be well-defined in general. It is very common in real analysis to consider the supremum of a set in replace of the intuitive 'largest' point. And this is the reason why it was mentioned that 'concepts in earlier sections such as supremum/infimum may be needed'. That was a hint to this question!

  There are some claims made by some students which are not correct or need justification. For example, it is not obvious why there must be an interval for which f is strictly increasing or decreasing (if f is not a constant function). Indeed, there exists a continuous but nowhere monotonic function! (You may search for these key words if you are interested to know more about this.) In general, be careful when you try to make some guess based on the intuitive understanding from the graph of functions.

- (Question 5a) The performance in this question is good. Note that we can actually prove that $\sum_{n=1}^{\infty} \sqrt{\frac{a_n^2+1}{2a_n}}$ diverges no matter the other series diverges or not. Indeed, the sequence

$\left\{\sqrt{\dfrac{a_n^2+1}{2a_n}}\right\}$ cannot converge to 0 since each term is always larger than or equal to 1.

- (Question 5b) I am surprised that so many students were not aware of the word 'nonnegative' when answering this question. This results in a wrong answer as a counterexample involving an alternating series does exist. I quite believe the performance would be much better if this mistake was not made, as the question is not hard at all.

  Among those attempted in the correct direction, one common mistake found is to mix up the sequence $\{b_n\}$ and the series $\sum_{n=1}^{\infty} b_n$. For example, one cannot simply say that if a sequence converges, then any of its subsequences must converge. This does not explain anything related to this question. In fact, the sequence of partial sums of $\sum_{k=1}^{\infty} b_{n_k}$ is not a subsequence of the sequence of partial sums of $\sum_{n=1}^{\infty} b_n$ in general.

- (Question 5c) In order to use the sequential criterion to find $\lim_{x\to 0} f(x)$, we need to show that $\lim_{n\to\infty} f(x_n)$ converges to the same limit for every sequence $\{x_n\}$ converging to 0 (with $x_n \neq 0$). In this question we only know the limit of one such sequence, namely $x_n = \dfrac{1}{n}$. So the sequential criterion cannot be used.

  If it is further given that $\lim_{x\to 0} f(x)$ exists, then the limit must be 0 by the sequential criterion. But in general the limit may not exist, and it is not hard to construct a counterexample based on this observation.